

Anany Singh

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EDUCATION

Indraprastha Institute of Information Technology

BTech in Electronics and VLSI Engineering

New Delhi, India

Aug 2023 – Present

EXPERIENCE

Software Development Engineer, Aakaar AI

May 2026 - Present

- Architected and built an AI-powered Listing Agent that generates optimized product imagery for Amazon sellers, supporting both ASIN-based and description-based (no-ASIN) entry flows to produce a unified set of 7 named slot images and 6 discovery/archetype variations per session.
- Designed the underlying data model and async job pipeline (sessions, generations, jobs, generated images) to handle status tracking, per-slot versioning, and free-tier quota enforcement, enabling reliable end-to-end generation and syncing of final images back to Amazon listings.
- Implemented user flow tracking and analytics instrumentation across the platform while contributing to broader UI/UX enhancements, improving usability and giving the product team clearer visibility into user behavior.

Undergraduate Researcher, MIDAS Lab

Jan 2026 - Present

- Contributing to ongoing research focused on evaluating and enhancing Code LLMs through the design and orchestration of autonomous agent-based workflows.
- Building evaluation pipelines and benchmarking frameworks to assess model performance, robustness, and edge-case behavior while maintaining strict confidentiality of proprietary methodologies.
- Collaborating with the research team to iterate on experimental setups, optimize agent coordination strategies, and accelerate insights into model capabilities and limitations.

Full-Stack Intern, Camarin AI

December 2024

- Implemented the existing backend model into a functional full-stack application, ensuring seamless integration between data pipelines and user interface components.
- Developed the entire frontend of the website using Next.js, focusing on responsive design and performance optimization to enhance user experience across devices.
- Collaborated closely with the development team to deliver key features ahead of schedule, accelerating the project timeline by 25 percent and improving frontend load times by 40 percent.

PROJECTS

Collaborative Drawing

May 2025 - May 2025

Tech Stack : *Typescript, PostgreSQL, React, TailwindCSS, Websockets* | [GitHub](#)

- Built a real-time collaborative drawing tool using WebSockets and Pub/Sub architecture, enabling sub-100ms latency updates for up to 50 concurrent users.
- Utilized p5.js to implement dynamic shape rendering—including squares, rectangles, and triangles—enabling real-time user interaction with shape parameters via numeric input, resulting in over 95% input responsiveness and below 50ms rendering delay.

SignTrack

March 2025 - March 2025

Tech Stack : *Python, OpenCV, Tensorflow, React* | [GitHub](#)

- Developed a traffic signal detection tool that processes dashcam footage to count the number of traffic signals encountered during a drive.
- Implemented object detection using a pretrained YOLOv8 model fine-tuned for traffic light recognition in diverse lighting and weather conditions.
- Achieved 91% detection accuracy across urban and highway datasets while optimizing inference speed for real-time use.

- Designed and implemented a novel Byzantine fault-tolerant clock synchronization protocol that combines the Brooks-Iyengar sensor fusion algorithm with Inter-Tertile Range (ITR) outlier detection to achieve bounded-error time agreement in adversarial peer-to-peer networks, tolerating up to $N/3-1$ malicious nodes.
- Engineered a switch from quartile-based to tertile-based outlier filtering to raise Byzantine tolerance from 25% to the theoretical 33% bound, then applied Brooks-Iyengar interval fusion (originally designed for physical sensor arrays) to aggregate the resulting uncertainty ranges into a single consensus offset with a quantified confidence interval.
- Built a modular 4-crate Rust workspace (core algorithms, libp2p-based networking, a multi-node Byzantine simulation framework, and a reference CLI node) and validated it across happy-path, single-Byzantine, and adversarial stress scenarios, maintaining consensus with only $\sim 100\text{ms}$ divergence at 50 nodes with 32% Byzantine peers.

RESEARCH PUBLICATIONS

Obey, Diverge, Collapse: Blind Obedience to Incorrect Instructions Drives Code LLMs to Irrecoverable Code Semantic Collapse 2026

arXiv preprint

arXiv:2607.04537

Detection of Fake News

July 2022

International Journal of All Research Education and Scientific Methods (IJARESM)

DOI : ISSN: 2455-6211

TECHNICAL SKILLS

Languages and Frameworks: C, C++, Python, Java, HTML, CSS, JavaScript, SQL, React, Node.js, Golang, Rust, Solidity, Redis, Monorepo, Turborepo, NextAuth, NextJs

Expertise Area: Machine Learning, Data Structures, Algorithms, Object-Oriented Design, Operating Systems, Database Management Systems

Tools and Technologies: Git, TensorFlow, PyTorch, Keras, Scikit learn, Tableau, Apache Servers, Pandas, NumPy, Postman, Docker, Google Cloud

LEADERSHIP / EXTRACURRICULAR

- Rated **1397** on **codeforces**
- Ranked among **Top 500** in the **National Mathematics Talent Contest** with over 60,000 registrations

DECLARATION

The above information is correct to the best of the knowledge. Any changes shall be informed to the concerned authorities.

Name: Anany Singh